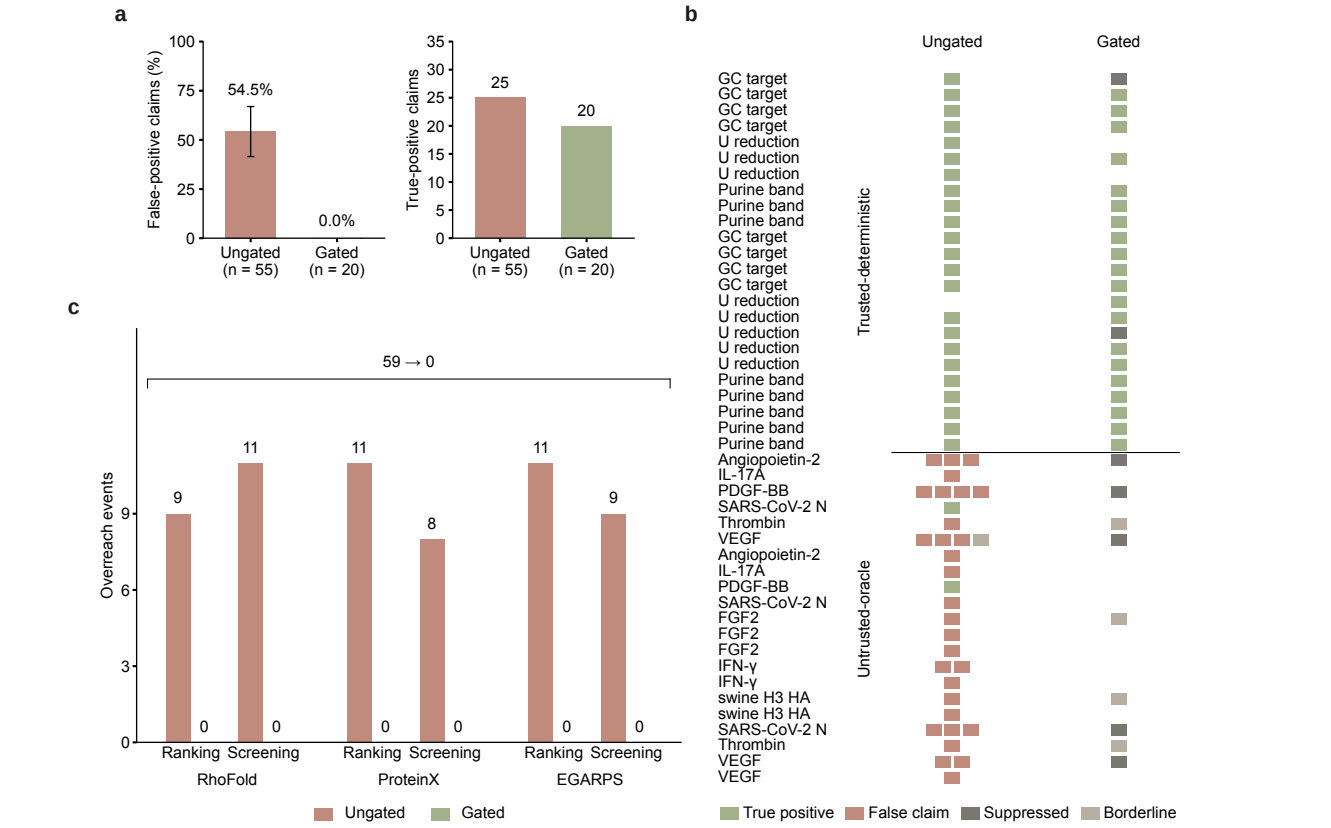
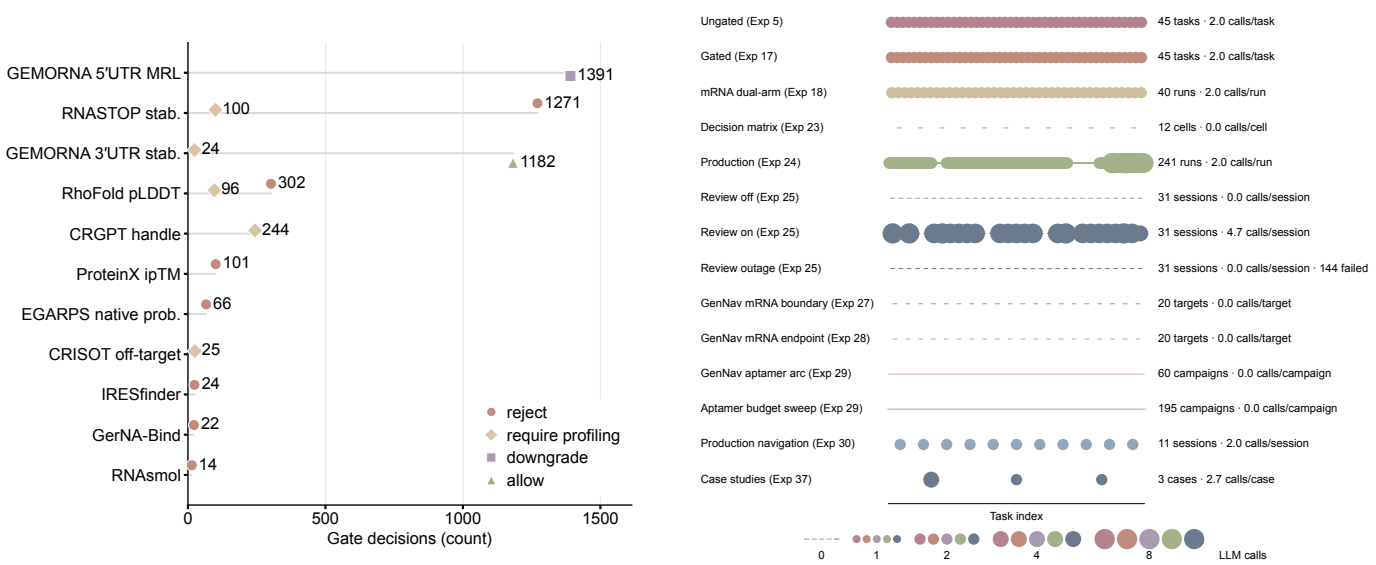
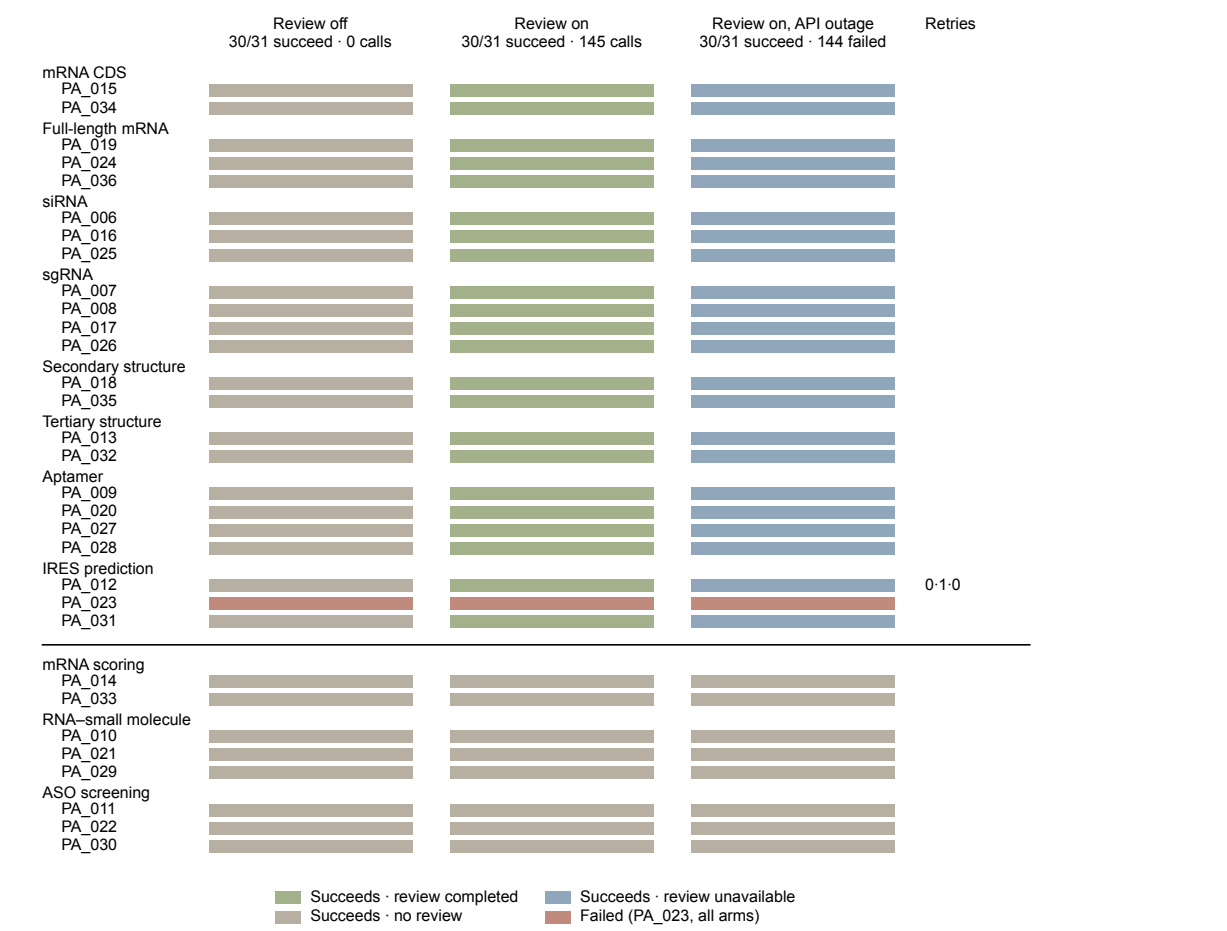

a
False-positive claims (%)
54.5%
0.0%
Ungated (n = 55)
Gated (n = 20)
True-positive claims
25
20
b
Ungated
Gated
Trusted-deterministic
Untrusted-oracle
GC target
U reduction
Purine band
Angiopoietin-2
IL-17A
PDGF-BB
SARS-CoV-2 N
Thrombin
VEGF
FGF2
IFN-γ
swine H3 HA
True positive
False claim
Suppressed
Borderline
c
59 → 0
Overreach events
Ranking
Screening
RhoFold
ProteinX
EGARPS
Ungated
Gated
d
Ungated
Gated
CAI report
MFE report
Constraints
Relative comparison
3′UTR stability rank
Cross-family MRL
Novel 5′UTR 7
Novel 5′UTR 11
Novel 5′UTR 13
Junction quality
MRL optimum
True positive
Neutral
No published claim
False claim
Suppressed
e
allow · pass n = 4
downgrade n = 2
blocked n = 3
DecisionGate (pre-action)
5′UTR MRL
3′UTR
certified · EGFP random-50
certified · var-len random
certified · var-len human
never-profiled family
certified family
never-profiled family
executed · MRL 6.28 = Exp-14
not blocked
not blocked
executed · MRL 6.32
blocked pre-call
blocked pre-call
ClaimGate (pre-publication)
5′UTR claim · certified
5′UTR claim · never-profiled
3′UTR claim · never-profiled
absolute scope stands
scope → within-target relative
no publishable basis
allow / pass
downgrade
require-profiling
reject


f
GEMORNA 5′UTR MRL
RNASTOP stab.
GEMORNA 3′UTR stab.
RhoFold pLDDT
CRGPT handle
ProteinX ipTM
EGARPS native prob.
CRISOT off-target
IRESfinder
GerNA-Bind
RNAsmol
reject
require profiling
downgrade
allow
Gate decisions (count)
h
Ungated (Exp 5)
Gated (Exp 17)
mRNA dual-arm (Exp 18)
Decision matrix (Exp 23)
Production (Exp 24)
Review off (Exp 25)
Review on (Exp 25)
Review outage (Exp 25)
GenNav mRNA boundary (Exp 27)
GenNav mRNA endpoint (Exp 28)
GenNav aptamer arc (Exp 29)
Aptamer budget sweep (Exp 29)
Production navigation (Exp 30)
Case studies (Exp 37)
Task index
LLM calls


g
Review off
30/31 succeed · 0 calls
Review on
30/31 succeed · 145 calls
Review on, API outage
30/31 succeed · 144 failed
Retries
mRNA CDS
Full-length mRNA
siRNA
sgRNA
Secondary structure
Tertiary structure
Aptamer
IRES prediction
mRNA scoring
RNA–small molecule
ASO screening
0·1·0
Succeeds · review completed
Succeeds · no review
Succeeds · review unavailable
Failed (PA_023, all arms)